

GIRRAWEE HIGH SCHOOL

MATHEMATICS

YEAR 11

MARCH 2003

TIME: 80 minutes

TEST 1

Instructions:

- Attempt all questions.
- Start each question on a separate page.
- All necessary working must be shown.
- Marks will be deducted for careless or badly arranged work.

Question 1 (12 marks)

- (a) Calculate $\frac{\sqrt{6.1+18.7}}{12.4}$ correct to 1 decimal place. 2
- (b) Find $\sqrt[3]{102}$ correct to 1 decimal place. 2
- (c) Write in scientific notation:
- (i) 2 100 000 000 2
- (ii) 0.00104 2
- (d) Find $[4.3 \times 10^{19}] \times [7.1 \times 10^{28}]$. Write your answer in scientific notation. 2
- (e) Express each of these recurring decimals as a fraction in its simplest form:
- (i) $0.5\ddot{8}$ 2
- (ii) $0.72\dot{6}$ 2

Question 2 (16 marks)

- (a) Simplify $4(a - 2) - 3(a - 5)$. 2
- (b) Expand and simplify:
- (i) $x^2 + 2x + x(x - 5)$ 2
- (ii) $2m^2(3m^3 + 2m + 6) - 5m(3m^2 - 4m)$ 2
- (c) Expand and simplify:
- (i) $(p^2 - q^2)(p + q)$ 2
- (ii) $(3x - 4)(x^2 - 2x + 5)$ 2
- (d) Expand and simplify $(3x - 4)(x - 2)(x - 2)$. 2
- (e) If $V = 3x^2y$, find V if $x = 0.2$ and $y = 75$. 2
- (f) If $K = (a + b)^3$, find K if $a = \frac{1}{5}$ and $b = 4\frac{1}{2}$. 2

Question 3 (23 marks)

(a) Write these surds in their simplest form:

(i) $\sqrt{250}$ 2

(ii) $\sqrt{\frac{9}{4}}$ 2

(b) Simplify:

(i) $\sqrt{8} + \sqrt{32}$ 2

(ii) $\sqrt{180} + \sqrt{128} - 2\sqrt{5}$ 2

(c) Find the value of a if $\sqrt{90} - \sqrt{40} = a\sqrt{10}$. 2

(d) Expand and simplify:

(i) $(\sqrt{3} + 2\sqrt{5})(\sqrt{3} - 2\sqrt{5})$ 2

(ii) $(\sqrt{3} + 2\sqrt{2})^2$ 2

(e) Rationalise the denominator and simplify:

(i) $\frac{2\sqrt{3}}{3\sqrt{2}}$ 2

(ii) $\frac{4}{\sqrt{3}+1}$ 2

(iii) $\frac{\sqrt{2}+1}{\sqrt{2}-1}$ 2

(f) Show $\frac{1}{5-\sqrt{7}} + \frac{1}{5+\sqrt{7}}$ is a rational number. 3

Question 4 (23 marks)

(a) Factorise:

(i) $3x + 3y + x^2 + xy$ 2

(ii) $(a + b)^2 + x(a + b)$ 2

(b) Factorise:

(i) $p^2 - 100$ 2

(ii) $12 - 3x^2$ 2

(c) Factorise:

(i) $x^2 - x - 12$ 2

(ii) $3x^2 - 10x + 8$ 2

(iii) $2 + x - 10x^2$ 2

(d) Simplify:

(i) $\frac{3x^2y}{6xy^2}$ 2

(ii) $\frac{x^2 - 9}{x - 3}$ 2

(e) Expand and simplify $(3a^2 + a)(a^3 - 3a^2 + 6a - 3)$. 3

(f) If $7 \cdot 2x^3 = 50$, find x to 1 decimal place. 2

Question 5 (10 marks)

(a) Simplify:

(i) $\frac{x+1}{2} + \frac{x}{3}$ 2

(ii) $\frac{1}{x} + \frac{1}{x+y}$ 2

(iii) $\frac{2}{(x+1)(x+3)} + \frac{1}{x+3}$ 2

(b) Factorise $8x^3 - 125$. 2

(c) If $T = a + (n-1)d$, find n if $T = 45$, $a = 5$ and $d = 2$. 2

Question 6 (12 marks)

(a) Simplify:

(i) $\frac{m^2 + 2m - 8}{m^2 - 4}$ 2

(ii) $\frac{1-t^2}{t-1}$ 2

(iii) $\frac{x^3 + 8}{x^2 - 2x + 4}$ 2

(b) Simplify:

(i) $\frac{3b-3c}{bc} \times \frac{b^2}{b^2-bc}$ 3

(ii) $\frac{x+1}{x+3} \div \frac{x^2-16}{x^2-x-12}$ 3

Question 1

a) 0.4 (to 1 d.p.)

②

b) $\sqrt[5]{102} = 2.5$

②

c) i) 2.1×10^9

①

ii) 1.04×10^{-3}

①

d) 3.053×10^{48}

②

e) i) $0.\dot{5}8$

Let $x = 0.5858$

$100x = 58.5858$

$99x = 58$

$\therefore 0.\dot{5}8 = \frac{58}{99}$

②

ii) $0.72\dot{6}$

Let $x = 0.72666$

$10x = 7.26666$

$9x = 6.54$

$x = \frac{6.54}{9} = \frac{654}{900}$

$\therefore 0.72\dot{6} = \frac{109}{150}$

②

Question 2

a) $4(a-2) - 3(a-5)$

$= 4a - 8 - 3a + 15$

$= a + 7$

②

b) i) $x^2 + 2x + x(x-5)$

$= x^2 + 2x + x^2 - 5x$

$= 2x^2 - 3x$

②

ii) $2m^2(3m^3 + 2m + 6) - 5m(3m^2 - 4m)$
 $= 6m^5 + 4m^3 + 12m^2 - 15m^3 + 20m^2$
 $= 6m^5 - 11m^3 + 32m^2$

②

c) i) $(p^2 - q^2)(p + q)$

$= p^3 + p^2q - pq^2 - q^3$

②

ii) $(3x-4)(x^2-2x+5)$

$= 3x^3 - 6x^2 + 15x - 4x^2 + 8x - 20$

$= 3x^3 - 10x^2 + 23x - 20$

②

d) $(3x-4)(x-2)(x-2)$

$= (3x-4)(x^2-4x+4)$

$= 3x^3 - 12x^2 + 12x - 4x^2 + 16x - 16$

$= 3x^3 - 16x^2 + 28x - 16$

②

e) $V = 3x^2y$

$V = 3 \times (0.2)^2 \times 75$

$V = 9$

②

f) $k = (a+b)^3$

$= \left(\frac{1}{5} + \frac{9}{2}\right)^3$

$= 103.823$

②

Question 3

$$\begin{aligned} \text{a) i) } \sqrt{250} &= \sqrt{25 \times 10} \\ &= 5\sqrt{10} \end{aligned}$$

②

$$\text{ii) } \sqrt{\frac{9}{4}} = \frac{3}{2}$$

②

$$\begin{aligned} \text{b) i) } \sqrt{8} + \sqrt{32} \\ &= 2\sqrt{2} + 4\sqrt{2} \\ &= 6\sqrt{2} \end{aligned}$$

②

$$\begin{aligned} \text{ii) } \sqrt{180} + \sqrt{128} - 2\sqrt{5} \\ &= \sqrt{36 \times 5} + \sqrt{64 \times 2} - 2\sqrt{5} \\ &= 6\sqrt{5} + 8\sqrt{2} - 2\sqrt{5} \\ &= 4\sqrt{5} + 8\sqrt{2} \end{aligned}$$

②

$$\begin{aligned} \text{c) } \sqrt{90} - \sqrt{40} &= a\sqrt{10} \\ 3\sqrt{10} - 2\sqrt{10} &= 1\sqrt{10} \\ \therefore a &= 1 \end{aligned}$$

②

$$\begin{aligned} \text{d) } (\sqrt{3} + 2\sqrt{5})(\sqrt{3} - 2\sqrt{5}) \\ &= 3 - 20 \\ &= -17 \end{aligned}$$

②

$$\begin{aligned} \text{ii) } (\sqrt{3} + 2\sqrt{2})^2 \\ &= 3 + 4\sqrt{6} + 8 \\ &= 11 + 4\sqrt{6} \end{aligned}$$

②

$$\begin{aligned} \text{e) i) } \frac{2\sqrt{3}}{3\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{2\sqrt{6}}{6} \\ &= \frac{\sqrt{6}}{3} \end{aligned}$$

②

$$\text{ii) } \frac{4}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$$

$$= \frac{4\sqrt{3}-4}{3-1}$$

$$= 2\sqrt{3}-2$$

②

$$\text{iii) } \frac{(\sqrt{2}+1)}{(\sqrt{2}-1)} \times \frac{(\sqrt{2}+1)}{(\sqrt{2}+1)}$$

$$= \frac{2+2\sqrt{2}+1}{2-1}$$

$$= 3+2\sqrt{2}$$

②

$$\text{f) } \frac{1}{5-\sqrt{7}} + \frac{1}{5+\sqrt{7}}$$

$$= \frac{5+\sqrt{7}+5-\sqrt{7}}{25-7}$$

$$= \frac{10}{18}$$

$$= \frac{5}{9} \text{ a rational number}$$

③

Question 4

$$\begin{aligned} \text{a) i) } 3x+3y+x^2+xy \\ &= 3(x+y) + x(x+y) \\ &= (3+x)(x+y) \end{aligned}$$

②

$$\begin{aligned} \text{ii) } (a+b)^2 + x(a+b) \\ &= (a+b)(a+b+x) \end{aligned}$$

②

$$H) i) p^2 - 100 = (p+10)(p-10) \quad (2)$$

$$ii) 12 - 3x^2 = 3(4 - x^2) \\ = 3(2-x)(2+x) \quad (2)$$

$$c) i) x^2 - x - 12 = (x-4)(x+3) \quad (2)$$

$$ii) 3x^2 - 10x + 8 = (3x-4)(x-2) \quad (2)$$

$$iii) 2 + x - 10x^2 = (2+5x)(1-2x) \quad (2)$$

$$d) i) \frac{3x^2y}{6xy^2} = \frac{x}{2y} \quad (2)$$

$$ii) \frac{x^2-9}{x-3} \cdot \frac{(x+3)\cancel{(x-3)}}{\cancel{(x-3)}} \\ = x+3 \quad (2)$$

$$e) (3a^2+a)(a^3-3a^2+6a-3) \\ = 3a^5 - 9a^4 + 18a^3 - 9a^2 + a^4 - 3a^3 + 6a^2 - 3a \\ = 3a^5 - 8a^4 + 15a^3 - 3a^2 - 3a \quad (3)$$

$$f) 7.2x^3 = 50 \\ x^3 = \frac{50}{7.2}$$

$$x = 2.6 \quad (2)$$

Question 5

$$a) i) \frac{x+1}{2} + \frac{x}{3} \\ = \frac{3x+3+2x}{6}$$

$$= \frac{5x+3}{6} \quad (2)$$

$$ii) \frac{1}{x} + \frac{1}{x+y}$$

$$= \frac{x+y+x}{x(x+y)}$$

$$= \frac{2x+y}{x(x+y)} \quad (2)$$

$$iii) \frac{2}{(x+1)(x+3)} + \frac{1}{x+3} \\ = \frac{2 + 1(x+1)}{(x+1)(x+3)}$$

$$= \frac{3+x}{(x+1)(x+3)} \quad (2)$$

$$b) 8x^3 - 125 = (2x)^3 - 5^3 \\ = (2x-5)(4x^2+10x+25) \quad (2)$$

$$c) T = a + (n-1)d$$

$$45 = 5 + (n-1)2$$

$$40 = 2n - 2$$

$$42 = 2n$$

$$n = 21$$

(2)

Question 6

$$a) i) \frac{m^2 + 2m - 8}{m^2 - 4}$$

$$= \frac{(m-2)(m+4)}{(m-2)(m+2)}$$

$$= \frac{m+4}{m+2}$$

②

$$ii) \frac{1-t^2}{t-1}$$

$$= \frac{(1+t)(1-t)}{-(1-t)}$$

$$= -(1+t)$$

②

$$iii) \frac{x^3 + 8}{x^2 - 2x + 4}$$

$$= \frac{(x+2)(x^2 - 2x + 4)}{(x^2 - 2x + 4)}$$

$$= x+2$$

③

$$b) i) \frac{3b-3c}{bc} \times \frac{b^2}{b^2-bc}$$

$$= \frac{3(b-c)}{bc} \times \frac{b^2}{b(b-c)}$$

$$= \frac{3}{c}$$

②

$$ii) \frac{x+1}{x+3} \div \frac{x^2-16}{x^2-x-12}$$

$$= \frac{x+1}{x+3} \div \frac{(x+4)(x-4)}{(x+3)(x-4)}$$

$$= \frac{x+1}{(x+3)} \times \frac{(x+3)(x-4)}{(x+4)(x-4)}$$

$$= \frac{x+1}{x+4}$$

③